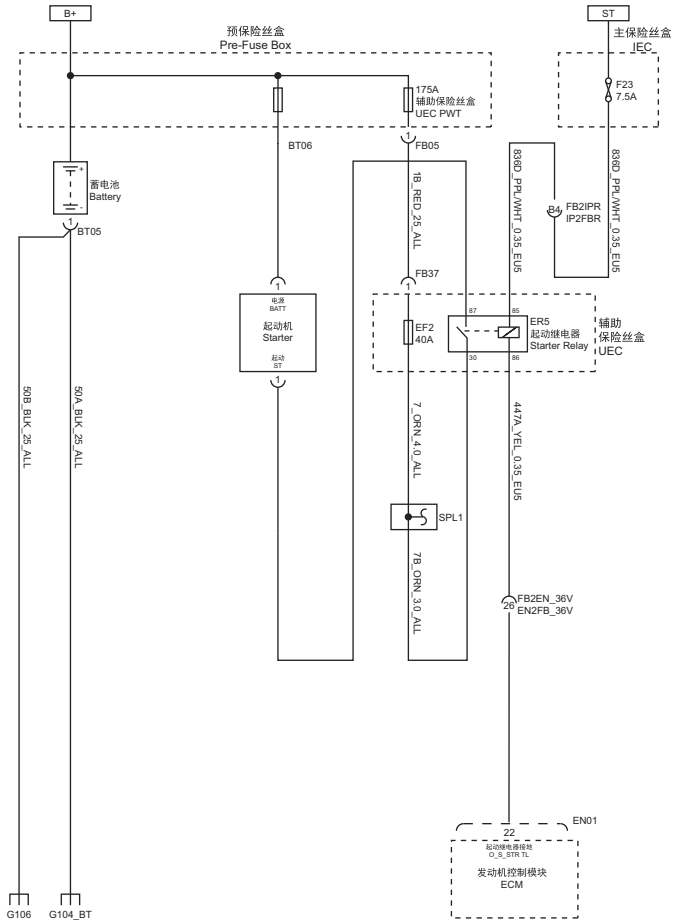
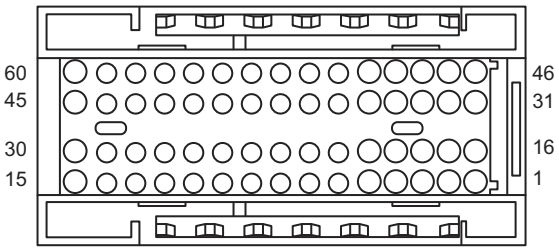
P0615、 P26E4

Description of fault diagnosis codes
P0615-13：No load error for starter
P26E4-11：Short circuit to ground error for low side of starter output
P26E4-12：Short circuit to battery error for low side of starter output
P26E4-4B：Over temperature error on ECU powerstage for low side of starter output
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • Starter relay fault • Engine control module fault

Circuit diagram



EN01



SV615051

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of engine control module for damage, poor contact, aging and looseness. 4 Check whether the result is normal.
		No → Repair or replace the faulted parts.
No ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 Is there any other fault code except for P0615,P26E4?
		Yes → Reference:DTC Summary List (EU5).
No ↓		
Step 3	Check the power supply circuit of starter relay.	
		1 Turn the ignition switch to OFF. 2 Unplug the starter relay. 3 Turn the ignition switch to START position. 4 Check the voltage between the terminal of starter relay and ground. Measure circuit: the voltage between terminal 85 of starter relay and ground. **Standard value: 10~14V** 5 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		

Step 4	Check the control circuit of starter relay (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Check the resistance between the terminal of starter relay and engine control module. Measure circuit: the resistance between terminal 86 of ER5 relay and EN01 plug terminal 22. **Standard value: ≈ 0Ω** 4 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check the control circuit of starter relay (short circuit).
	1 Check the resistance between the harness plug of engine control module and ground. Measure circuit: the resistance between EN01 plug terminal 22 and ground. **Standard value: ∞** 2 Check the voltage between the harness plug of engine control module and ground. Measure circuit: the voltage between EN01 plug terminal 22 and ground. **Standard value: ≈ 0V** 3 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 6	Check the starter relay(components).
	1 Apply 12V voltage to terminals 85 and 86 of the relay. 2 Measure the resistance among the terminals of relay. Measuring circuit: the resistance between terminal 30 and terminal 87 of the relay. **Standard value: ≈ 0Ω** 3 Check whether the result is normal?
	No → Replace the starter relay.
Next ↓	
Step 7	Complete the check and confirm whether the system works normally nor not.